

GLOBAL HEALTH ANALYTICS

COVID-19 Global Impact

Dashboard & Strategic Briefing

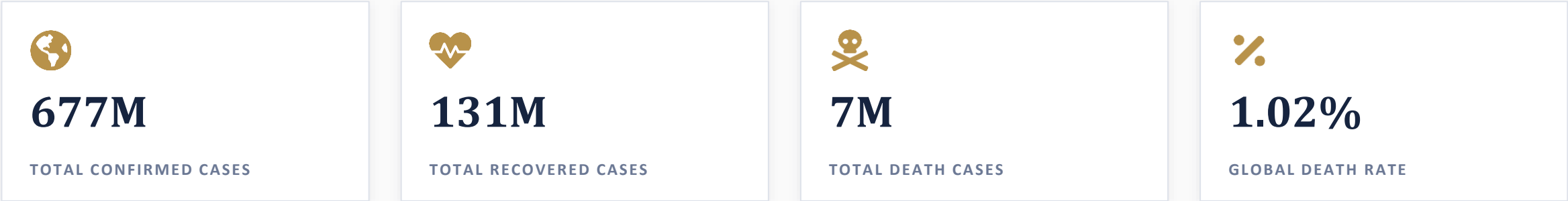
A data-driven review of confirmed cases, deaths, and recoveries across 289 countries and territories, January 2020 through March 2023.

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AnalystLab Africa — Data Analytics Internship, Week 4

Dataset & Key Performance Indicators

Source: Johns Hopkins University CSSE COVID-19 Time Series (GitHub) · Confirmed, Death, and Recovery datasets merged by country and date



DATASET SCOPE

- 289 countries and regions tracked
- 1,143 daily records spanning Jan 2020 – Mar 2023
- 3 linked datasets unified into one model

BUSINESS PROBLEM

Decision-makers need a clear, interactive view of how COVID-19 spread globally over time — identifying which countries were hit hardest, how recovery and death rates compared, and what patterns can inform future public health response.

Interactive Power BI Dashboard

Fully interactive with country, date, and province/state filters applied across all visuals



Date

All

Country

All

Province/State

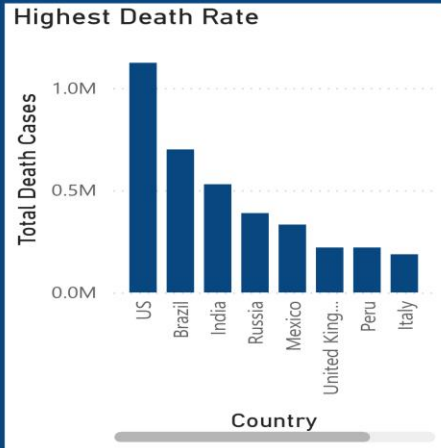
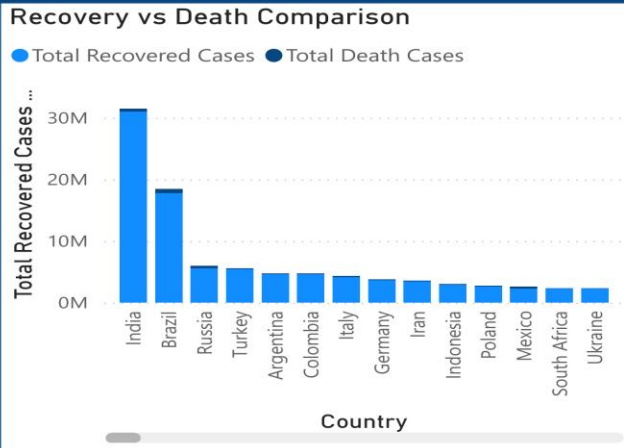
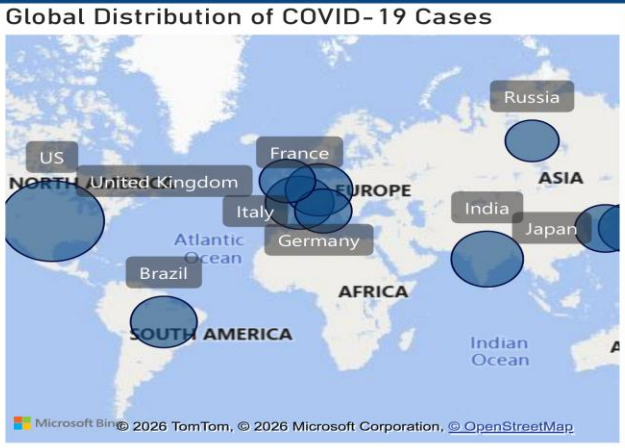
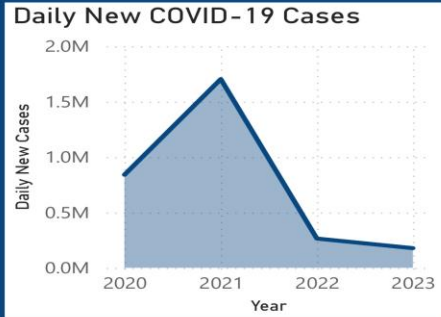
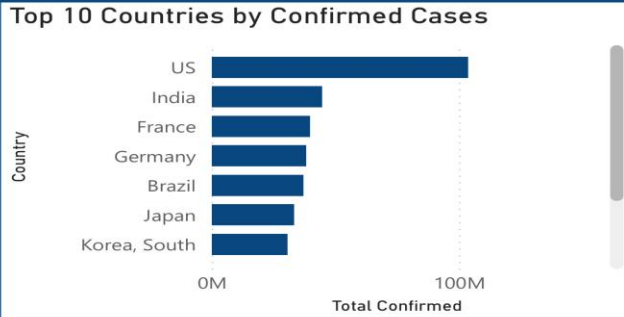
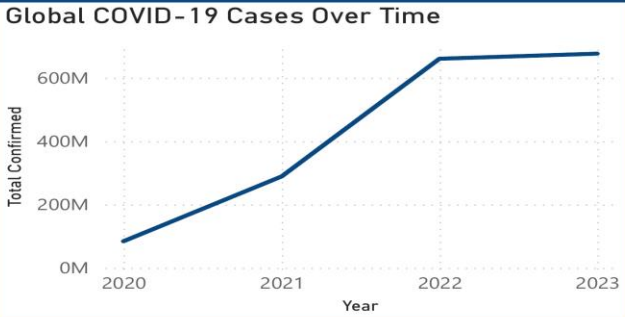
All

Total Confirmed
677M

Total Recovered Cases
131M

Total Death Cases
7M

Death Rate
1.02%



Key Insights from the Data



Growth Pattern

Global confirmed cases rose sharply through 2020–2021, then plateaued from 2022 onward as vaccination and immunity expanded — the curve flattens rather than continuing to climb.



Peak Transmission

Daily new cases peaked around 2021, then declined steadily into 2022–2023, reflecting the impact of vaccines, treatments, and public health measures over time.



Country Concentration

The US, India, France, Germany, and Brazil account for the largest share of confirmed cases, showing impact was concentrated in a handful of large, high-mobility nations.



Death Rate Gap

The global death rate sits at 1.02%, but countries like the US, Brazil, and India show disproportionately higher absolute death tolls — pointing to healthcare capacity as a key factor.

Recommendations & Conclusion

RECOMMENDATIONS

- Prioritize early surveillance in high-mobility countries, where case growth historically accelerated fastest.
- Strengthen healthcare capacity in countries showing high death tolls relative to confirmed cases.
- Use daily new case trends, not just cumulative totals, to time public health interventions.

CHALLENGES FACED

- Raw data had dates stored as columns instead of rows, requiring an unpivot transformation in Power Query.
- Resolving DAX measure and data-type errors while merging the three datasets (confirmed, deaths, recovered).
- Balancing visual density to keep the dashboard clean while showing six chart types.

CONCLUSION

This dashboard turns three years of raw Johns Hopkins COVID-19 data into a clear, interactive story — showing how the pandemic grew, peaked, and stabilized globally, and which countries bore the greatest burden. It demonstrates how Power BI can transform large time-series datasets into insights decision-makers can act on quickly.